masses of matter is one of the most challenging," Wheeler wrote soon thereafter; and he set out to complete the exploration of stellar graveyards that Chandrasekhar and Oppenheimer and Volkoff had begun.

To make his task very precise, Wheeler formulated a careful characterization of the kind of matter from which cold, dead stars should be made: He called it *matter at the endpoint of thermonuclear evolution,* since the word *thermonuclear* had become popular for the fusion reactions that power nuclear burning in stars and also power the hydrogen bomb. Such matter would be absolutely cold, and it would have burned its nuclear fuel completely; there would be no way, by any kind of nuclear reaction, to extract any more energy from the matter's nuclei. For this reason, the nickname *cold, dead matter* will be used in this book instead of "matter at the endpoint of thermonuclear evolution."

Wheeler set himself the goal to understand *all* objects that can be made from cold, dead matter. These would include small objects like balls of iron, heavier objects such as cold, dead planets made of iron, and still heavier objects: white dwarfs, neutron stars, and whatever other kinds of cold, dead objects the laws of physics allow. Wheeler wanted a *comprehensive* catalog of cold, dead things.